

apply a passive high thermal conductivity coating on the leading edge that spreads impact generated heat, reducing localized thermal spikes that accelerate rain erosion

Executive Technology Brief

Domain: thermal

Technology Maturity: ENGINEERING_DEFINITION

Transfer Posture: Sponsored Validation

Problem: Documented problem 'leading-edge erosion' concerning offshore wind turbine blade. User-submitted engineering problem (session input): "Why do leading-edge erosion of offshore wind turbine blades degrade annual energy production, and can a passive surface design extend blade life in rain erosion?". ROUTING: nhtsa_complaints: not queried — no vehicle (make|model|year) extracted from the problem; this source answers vehicle-parameterized questions only. [nhtsa_complaints] retrieval status NOT_QUERIED_GRAMMAR (no vehicle extracted; source grammar is make|model|year (routing decision, not absence)) — provider outcome recorded, not treated as absence (Art. XXI.3). [doe_osti] 10 records retrieved live (2025-2026). Illustrative records: 'Presentation on the safety considerations with regard to commercial fusion energy'; 'Microgrid implementation can help improve electrical service, reliability, and resilience for localized communities. However, they may be susceptible to damage from natural disasters and extreme weath'. LIMITS STAMPED INLINE: literature context establishes the problem class, not event rates. Problem-existence basis: live records retrieved.

Mechanism: thermal anomaly response (temperature variation due to surface damage)

Architecture: PROPOSED architecture (no physical system exists): heat source characterization; conduction path / spreader; reject interface; safety limit interlock — realizing: apply a passive high thermal conductivity coating on the leading edge that spreads impact generated heat, reducing localized thermal spikes that accelerate rain erosion

Governing Model: Thermal management / heat transfer governing relations proposed for this invention; SYMBOLIC ONLY until inputs are sourced

Key Equations:

TH-002: Convective heat transfer (Newton's law of cooling). Expression: $q = h * A * dT$. Applicability to this invention: APPLICABLE; holds under: steady state; h characterized for the actual flow geometry. Assumptions: h correlations are geometry- and regime-specific; using a correlation outside its validated range is forbidden. Engaged variables: dT.

TH-003: Series thermal resistance. Expression: $R_{total} = \sum(L_i / (k_i * A_i))$. Applicability to this invention: APPLICABLE; holds under: one-dimensional steady conduction through layers. Assumptions: interface contact resistances dominate at small scale and must be measured, not assumed zero. Engaged variables: R_{total} , k_i .

TH-001: Fourier conduction. Expression: $q = -k * A * dT/dx$. Applicability to this invention: APPLICABLE; holds under: steady state, isotropic conduction. Assumptions: steady-state conduction (no transient term); isotropic, homogeneous medium conductivity k; one-dimensional gradient (lateral gradients neglected). Engaged variables: k, dT/dx.

Kill Condition: killer experiment fails: falsification_test_from_candidate (decision scores: EIG 0.4557178723356231, EIG-per-cost 0.4557178723356231; the falsification test carries the pre-registered pass/fail before any transfer)

Current State: ENGINEERING_DEFINITION. No physical prototype. No clinical validation.

Disclosure: This dossier distinguishes established evidence from proposals and unresolved questions.